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Method and time synchronization (TS) node for enabling extended holdover time

Abstract

The present disclosure provides a method and time synchronization node for enabling extended holdover time. The method comprises: detecting whether the TS node loses time synchronization with its master TS node or not; and transmitting a first announce message comprising a first indicator to one or more slave TS nodes in response to detecting that the TS node loses the time synchronization with its master TS node, wherein the first indicator indicates a holdover time budget for enabling extended holdover time at the one or more slave TS nodes.

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Background/Summary

(1) This application is a 35 U.S.C. § 371 national phase filing of International Application No. PCT/CN2020/088776, filed May 6, 2020, the disclosure of which is incorporated herein by reference in its entirety.
TECHNICAL FIELD
(2) The present disclosure is related to the field of telecommunication, and in particular, to a method and time synchronization (TS) node for enabling extended holdover time.
BACKGROUND
(3) In modern computer networks, for example, telecommunication networks, time synchronization is critical because every aspect of operating, managing, securing, planning, and debugging a network involves determining when events happened and/or when events will or should happen. For example, an eNodeB (eNB) in a Long Term Evolution (LTE) network has to be synchronized and cooperated with other eNBs nearby to function properly. Otherwise significant interferences to the eNB will be caused by asynchronous transmissions from the other eNBs, and vice versa.
(4) One of the protocols for synchronization is the Precision Time Protocol (PTP), also known as Institute of Electrical and Electronics Engineers (IEEE) 1588, which is designed to synchronize clocks at nodes in a network to sub-microsecond precision, making it suitable for measurement and control systems. PTP is currently employed to synchronize financial transactions, mobile phone tower transmissions, sub-sea acoustic arrays, and networks that require precise timing.
(5) The original version of PTP, IEEE 1588-2002, was published in 2002. IEEE 1588-2008, also known as PTP Version 2 is not backward compatible with the original 2002 version. IEEE 1588-2019 was published in November 2019 and includes backward-compatible improvements to the 2008 publication. IEEE 1588-2008 (or PTP Version 2) is an industry-standard protocol that enables the precise transfer of phase and time to synchronize clocks over packet-based Ethernet networks. It synchronizes the local slave clock on each network device with a system Grandmaster clock and uses traffic time-stamping, with sub-nanoseconds granularity, to deliver the very high accuracy of synchronization needed to ensure the stability of base station frequency and handovers. Timestamps between master and slave devices are sent within specific PTP packets and in its basis form the protocol is administration-free.
(6) IEEE 1588-2008 includes a profile concept defining PTP operating parameters and options. Several profiles have been defined for applications including telecommunications, electric power distribution, and audiovisual. One of the profiles adopted by current telecommunication industry is the Telecommunication Standardization Sector of International Telecommunication Union (ITU-T) G.8275.1/Y.1369.1. The ITU-T G.8275.1 profile defines a specific architecture to allow the distribution of phase/time with full timing support network.
(7) However, there are still many unsolved problems in existing time synchronization mechanisms. For example, how to extend holdover time at a TS node while it still has the capability to do so is one of them.
SUMMARY
(8) According to an aspect of the present disclosure, a method at a time synchronization (TS) node for enabling extended holdover time is provided. The method comprises: detecting whether the TS node loses time synchronization with its master TS node or not; and transmitting a first announce message comprising a first indicator to one or more slave TS nodes in response to detecting that the

TS node loses the time synchronization with its master TS node, wherein the first indicator indicates a holdover time budget for enabling extended holdover time at the one or more slave TS nodes.
(9) In some embodiments, after an initial transmission of the first announce message, one or more further transmissions of the first announce message occur periodically with a value of the holdover time budget comprised in the first announce message being gradually reduced as the time elapses. In some embodiments, the method further comprises: entering into a "Holdover-In-Specification" state upon detecting that the TS node loses the time synchronization with its master TS node. In some embodiments, the first announce message further comprises a second indicator indicating that the TS node enters into the "Holdover-In-Specification" state. In some embodiments, the first indicator indicates an initial holdover time budget which is pre-configured at the TS node. In some embodiments, upon detecting that the TS node loses the time synchronization with its master TS node, the method further comprises: starting a holdover timer with its expiration time set as the initial holdover time budget.
(10) In some embodiments, before detecting that the TS node loses the time synchronization with its master TS node, the method further comprises: receiving a second announce message which comprises a third indicator, the third indicator indicating a holdover time budget of the master TS node when the second announce message was transmitted; and storing the holdover time budget of the master TS node. In some embodiments, the first indicator indicates the minimum of an initial holdover time budget which is pre-configured at the TS node and the stored holdover time budget of the master TS node if the master TS node belongs to a same TS region as that of the TS node. In some embodiments, upon detecting that the TS node loses the time synchronization with its master TS node, the method further comprises: starting a holdover timer with its expiration time set as the minimum of the initial holdover time budget which is pre-configured at the TS node and the stored holdover time budget of the master TS node.
(11) In some embodiments, the first indicator indicates the sum of an initial holdover time budget which is pre-configured at the TS node and the stored holdover time budget of the master TS node if the master TS node belongs to a TS region other than that of the TS node. In some embodiments, upon detecting that the TS node loses the time synchronization with its master TS node, the method further comprises: starting a holdover timer with its expiration time set as the sum of the initial holdover time budget which is pre-configured at the TS node and the stored holdover time budget of the master TS node.
(12) In some embodiments, the method further comprises: placing the TS node into a "Holdover-Out-Of-Specification" state upon the expiration of the holdover timer. In some embodiments, if multiple second announce messages are received from the master TS node, the step of storing the holdover time budget of the master TS node comprises: overwriting the previously stored holdover time budget of the master TS node with the holdover time budget of the master TS node indicated in the last received second announce message.
(13) In some embodiments, in response to detecting that time synchronization with the master TS node which is synchronized to a primary reference time source (PRTS) or in the "Holdover-In-Specification" state is restored, the method further comprises: placing the TS node into the "locked" state; stopping the holdover timer; and transmitting an announce message to the one or more slave TS nodes to inform the restoration of the time synchronization.
(14) In some embodiments, in response to detecting that time synchronization with the master TS node which is synchronized to a primary reference time source (PRTS) is restored, the method further comprises: clearing the stored holdover time budget of the master TS node. In some embodiments, in response to detecting that time synchronization with the master TS node which is in the "Holdover-In-Specification" state is restored, the method further comprises: updating the holdover time budget of the master TS node when receiving the second announce message.
(15) In some embodiments, each of the TS nodes is an IEEE 1588 Precision Time Protocol (PTP)-

compliant node. In some embodiments, each of the announce messages is a PTP-compliant ANNOUNCE message.
(16) According to another aspect of the present disclosure, a time synchronization (TS) node is provided. The TS node comprises: a processor; a memory storing instructions which, when executed by the processor, cause the processor to perform any method described above.
(17) According to yet another aspect of the present disclosure, a non-transitory computer readable storage medium is provided. The non-transitory computer readable storage medium stores instructions which, when executed by a processor, cause the processor to perform any method described above.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The foregoing and other features of the present disclosure will become more fully apparent from the following description and appended claims, taken in conjunction with the accompanying drawings. Understanding that these drawings depict only several embodiments in accordance with the disclosure and therefore are not to be considered limiting of its scope, the disclosure will be described with additional specificity and detail through use of the accompanying drawings.
(2) FIG. **1** is a diagram illustrating a telecommunication network comprising various functional entities in which an embodiment of the present disclosure may be applicable.
(3) FIG. **2**A-FIG. **2**C are schematic diagrams illustrating a TS network comprising multiple TS nodes according to an embodiment of the present disclosure.
(4) FIG. **3**A-FIG. **3**G are schematic diagrams illustrating another TS network comprising multiple TS nodes according to another embodiment of the present disclosure.
(5) FIG. **4**A-FIG. **4**F are schematic diagrams illustrating yet another TS network comprising multiple TS nodes according to yet another embodiment of the present disclosure.
(6) FIG. **5** is a flow chart of an exemplary method for enabling extended holdover time according to an embodiment of the present disclosure.
(7) FIG. **6** schematically shows an embodiment of an arrangement which may be used in a TS node according to an embodiment of the present disclosure.
DETAILED DESCRIPTION
(8) Hereinafter, the present disclosure is described with reference to embodiments shown in the attached drawings. However, it is to be understood that those descriptions are just provided for illustrative purpose, rather than limiting the present disclosure. Further, in the following, descriptions of known structures and techniques are omitted so as not to unnecessarily obscure the concept of the present disclosure.
(9) Those skilled in the art will appreciate that the term “exemplary” is used herein to mean “illustrative,” or “serving as an example,” and is not intended to imply that a particular embodiment is preferred over another or that a particular feature is essential. Likewise, the terms “first” and “second,” and similar terms, are used simply to distinguish one particular instance of an item or feature from another, and do not indicate a particular order or arrangement, unless the context clearly indicates otherwise. Further, the term “step,” as used herein, is meant to be synonymous with “operation” or “action.” Any description herein of a sequence of steps does not imply that these operations must be carried out in a particular order, or even that these operations are carried out in any order at all, unless the context or the details of the described operation clearly indicates otherwise.
(10) Conditional language used herein, such as “can,” “might,” “may,” “e.g.,” and the like, unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain embodiments include, while other embodiments do not include,

certain features, elements and/or states. Thus, such conditional language is not generally intended to imply that features, elements and/or states are in any way required for one or more embodiments or that one or more embodiments necessarily include logic for deciding, with or without author input or prompting, whether these features, elements and/or states are included or are to be performed in any particular embodiment. Also, the term “or” is used in its inclusive sense (and not in its exclusive sense) so that when used, for example, to connect a list of elements, the term “or” means one, some, or all of the elements in the list. Further, the term “each,” as used herein, in addition to having its ordinary meaning, can mean any subset of a set of elements to which the term “each” is applied.

(11) The term “based on” is to be read as “based at least in part on.” The term “one embodiment” and “an embodiment” are to be read as “at least one embodiment.” The term “another embodiment” is to be read as “at least one other embodiment.” Other definitions, explicit and implicit, may be included below. In addition, language such as the phrase “at least one of X, Y and Z,” unless specifically stated otherwise, is to be understood with the context as used in general to convey that an item, term, etc. may be either X, Y, or Z, or a combination thereof.

(12) The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limitation of example embodiments. As used herein, the singular forms “a”, “an”, and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises”, “comprising”, “has”, “having”, “includes” and/or “including”, when used herein, specify the presence of stated features, elements, and/or components etc., but do not preclude the presence or addition of one or more other features, elements, components and/or combinations thereof. It will be also understood that the terms “connect(s),” “connecting”, “connected”, etc. when used herein, just mean that there is an electrical or communicative connection between two elements and they can be connected either directly or indirectly, unless explicitly stated to the contrary.

(13) Of course, the present disclosure may be carried out in other specific ways than those set forth herein without departing from the scope and essential characteristics of the disclosure. One or more of the specific processes discussed below may be carried out in any electronic device comprising one or more appropriately configured processing circuits, which may in some embodiments be embodied in one or more application-specific integrated circuits (ASICs). In some embodiments, these processing circuits may comprise one or more microprocessors, microcontrollers, and/or digital signal processors programmed with appropriate software and/or firmware to carry out one or more of the operations described above, or variants thereof. In some embodiments, these processing circuits may comprise customized hardware to carry out one or more of the functions described above. The present embodiments are, therefore, to be considered in all respects as illustrative and not restrictive.

(14) Although multiple embodiments of the present disclosure will be illustrated in the accompanying Drawings and described in the following Detailed Description, it should be understood that the disclosure is not limited to the disclosed embodiments, but instead is also capable of numerous rearrangements, modifications, and substitutions without departing from the present disclosure that as will be set forth and defined within the claims.

(15) Further, please note that although the following description of some embodiments of the present disclosure is given in the context of Long Term Evolution (LTE), the present disclosure is not limited thereto. In fact, as long as time synchronization is involved, the inventive concept of the present disclosure may be applicable to any appropriate communication architecture, for example, to Global System for Mobile Communications (GSM)/General Packet Radio Service (GPRS), Enhanced Data Rates for GSM Evolution (EDGE), Code Division Multiple Access (CDMA), Wideband CDMA (WCDMA), Time Division-Synchronous CDMA (TD-SCDMA), CDMA2000, Worldwide Interoperability for Microwave Access (WiMAX), Wireless Fidelity (Wi-Fi), 5th Generation New Radio (5G NR), etc. Therefore, one skilled in the arts could readily understand

that the terms used herein may also refer to their equivalents in any other infrastructure. For example, the term "User Equipment" or "UE" used herein may refer to a mobile device, a mobile terminal, a mobile station, a user device, a user terminal, a wireless device, a wireless terminal, or any other equivalents. For another example, the term "eNodeB" or "eNB" used herein may refer to a base station, a base transceiver station, an access point, a hot spot, a NodeB, an Evolved NodeB, a gNB, a network element, or any other equivalents. Further, the term "node" used herein may refer to a UE, a functional entity, a network entity, a network element, a network equipment, a TS node, or any other equivalents.

(16) Further, although the embodiments described below are given in the context of PTP protocol and ITU-T G.8275.1 profile, the present disclosure is not limited thereto. In fact, the embodiments of the present disclosure may be applicable to any protocol and/or profile for time synchronization with appropriate modifications, adjustments, and/or changes which are known to one skilled in the art, if necessary.

(17) Before the description of the embodiments of the present disclosure is given in details, some definitions of the terms will be explained below to facilitate the understanding of the present disclosure.

(18) Node or TS node: A device that can issue or receive communications for time synchronization on a network.

(19) Clock: A node participating in a time synchronization process, such as Precision Time Protocol (PTP), that is capable of providing a measurement of the passage of time since a defined epoch.

(20) Domain: A logical grouping of clocks that synchronize to each other using the time synchronization process, but that are not necessarily synchronized to clocks in another domain.

(21) Region: A logical part of a domain which comprises no node from any other region in the domain.

(22) Primary reference time source: A source of time that is traceable to international standards, such as Global Positioning System (GPS).

(23) Master clock: In the context of a single communication path for time synchronization, a clock that is the source of time to which all other clocks on that path synchronize.

(24) Slave clock: A clock that is synchronized to a master clock.

(25) Grandmaster clock: Within a domain, a clock that is the ultimate source of time for clock synchronization using the time synchronization process.

(26) Port: A logical access point of a clock for time synchronization communications to the communications network.

(27) Master port: A port of a master clock via which a slave clock is synchronized or locked to the master clock

(28) Slave port: A port of a slave clock via which the slave clock is synchronized or locked to its master clock.

(29) Passive port: A port of a clock which does not place any messages on its communication path except for Pdelay_Req, Pdelay_Resp, Pdelay_Resp_Follow_Up, or signaling messages, or management messages that are required responses to another management message.

(30) Holdover: A clock previously synchronized to another clock (normally a primary reference or a master clock) but now free-running based on its own internal oscillator, whose frequency is being adjusted using data acquired while it had been synchronized to the other clock. It is said to be in holdover or in the holdover state, as long as it is within its accuracy requirements.

(31) Ordinary clock: A clock that has a single port in a domain and maintains the timescale used in the domain. It may serve as a source of time, i.e., be a master clock, or may synchronize to another clock, i.e., be a slave clock.

(32) Boundary clock: A clock that has multiple ports in a domain and maintains the timescale used in the domain. It may serve as the source of time, i.e., be a master clock, and may synchronize to another clock, i.e., be a slave clock.

(33) FIG. **1** is a diagram illustrating a telecommunication network **10** comprising various functional entities or nodes in which an embodiment of the present disclosure may be applicable. Please note that although FIG. **1** shows an exemplary telecommunication network **10** based on the 4G LTE technology, the present disclosure is not limited thereto as mentioned above.
(34) As shown in FIG. **1**, the telecommunication network **10** may comprise an access network **120** and a core network **130** as defined in 4G LTE. The access network **120** may comprise one or more eNodeBs (eNBs) (e.g., the eNB**1 125-1** and eNB**2 125-2**) for providing user equipments (UE) (e.g., UE**1 110-1** and UE**2 110-2**) with access to the core network **130** and finally to the Internet **160**.
(35) Further, the core network **130** may comprise a Serving-Gateway (S-GW) **131**, a Packet Data Network (PDN) gateway (P-GW) **132**, a Policy & Charging Rules Function (PCRF) **133**, a Mobility Management Entity (MME) **134**, a Home Subscriber Server (HSS) 135, and a Serving GPRS Support Node (SGSN) **136**, among other functional entities. All these nodes should be synchronized to each other, or the performance of the telecommunication network **10** will be seriously degraded.
(36) For example, if the eNB**1 125-1** is not synchronized to the eNB **2 125-2** directly via the X2 interface therebetween or via any other interface (e.g. **51**-U interfaces between the eNBs **125-1**/**125-2** and the S-GW **131**) indirectly, then wireless transmission from the eNB**2 125-2** to the UE**2 110-2** may cause a significant interference to the transmission from the eNB**1 125-1** to the UE**1 110-1**, and vice versa. For another example, if the S-GW **131** is not synchronized to the MME **134** via the S11 interface directly or via any other interface indirectly, then user plane data destined to the UE**1 110-1** may be incorrectly routed to the eNB**1 125-1** instead of eNB**2 125-2** when a handover of the UE**1 110-1** from the cell served by eNB**1 125-1** to the cell served by the eNB**2 125-2** is approved by the MME **134** while the S-GW **131** is not notified of such a handover due to the asynchronous operations between the S-GW **131** and the MME **134**. Based on these examples, it is obvious that it is very important to maintain the nodes in the telecommunication network **10** to be synchronized to each other.
(37) Next, a time synchronization mechanism based on PTP/ITU-T G.8275.1 will be explained with reference to FIG. **2**A-FIG. **2**C.
(38) FIG. **2**A-FIG. **2**C are schematic diagrams illustrating a TS network (or domain) **20** comprising multiple TS nodes according to an embodiment of the present disclosure. As shown in FIG. **2**A-FIG. **2**C, the TS network **20** may comprise one or more TS nodes, such as, a grand master clock (e.g. T-GM **200**), two boundary clocks (e.g. T-BC-1 **210** and T-BC-2 **220**), and two slave-only ordinary clocks (e.g. T-TSC-1 **222** and T-TSC-2 **224**), which are synchronized to each other. Further, there is also a primary reference time source (PRTS) **230** to which the T-GM **200** (and the whole TS network **20**) is synchronized, and it typically does not belong to the TS domain **20** according to the definition of "domain" above.
(39) As shown in FIG. **2**A-FIG. **2**C, these nodes **200**, **210**, **220**, **222**, and **224** in the TS domain **20** may be divided into two regions, a transport region and a radio region. In some embodiments, each of the nodes **200**, **210**, and **220** in the transport region may be the nodes in the core network **130**, such as S-GW **131**, MME **134**, P-GW **132**, etc., as shown in FIG. **1**, while the nodes in the radio region may be the nodes in the access network **120**, such as eNB**1 125-1**, eNB**2 125-2**, also as shown in FIG. **1**. Further, in the embodiment shown in FIG. **2**A, the T-TSC-1 **222** and T-TSC-2 **224** are slave clocks synchronized to the T-BC-2 **220**, as indicated by the reference symbols "S" (for slave port) and "M" (for master port). Further, the T-BC-1 **210** and T-BC-2 **220** are slave clocks synchronized to the T-GM **200**. Therefore, the T-TSC-1 **222** and T-TSC-2 **224** are also slave clocks synchronized to the T-GM **200**. Furthermore, the T-BC-1 **210** may also have an additional master port which may communicate with a passive port of the T-BC-2 **220** to exchange data for time synchronization, as shown in FIG. **2**A, which will be explained later.
(40) Please note that although FIG. **2**A-FIG. **2**C show only certain nodes and certain links between those nodes, the present disclosure is not limited thereto, and there may be additional nodes, less

nodes, different nodes, and/or additional links, less links or different links therebetween. For example, a link may be present between T-BC-1 **210** and T-TSC-1 **222** though it is not shown in FIG. **2**A-FIG. **2**C. For another example, an additional node such as a transparent clock T-TC may be present between the T-GM **200** and the T-BC-2 **220**. However, please note that, according to the provisions of subclause 6.7.1 of IEEE 1588-2008, "In all other cases, PTP assumes that the underlying bridging or routing protocols ensure that PTP message forwarding avoids loops." Therefore, no loop is present in the embodiments described herein. However, the present disclosure is not limited thereto.

(41) Referring to FIG. **2**A, the TS domain **20** is operated in a locked state. In other words, each node of the TS domain **20** is synchronized or locked to its master node, and ultimately to the grand master node T-GM **200**, which is synchronized or locked to its master node PRTS **230**, which could be an GPS satellite in the embodiment shown in FIG. **2**A.

(42) As mentioned in the Background section, a TS node may temporarily lose its time synchronization with its master node. As shown in FIG. **2**B, the nodes in the TS domain **20** other than the T-GM **200** lose their time synchronization with the grand master node T-GM **200**, for example, due to the power-down or link failure of the T-GM **200**. In such a case, the boundary clock node T-BC-1 **210** may become the new grand master clock, for example, based on the best master clock algorithm specified in PTP and ITU-T G.8275.1, and enters into the "Holdover-In-Specification" state, which indicates that it is in the holdover state and the time error at its output is still within the holdover specification. The T-BC-1 **210** may then transmit an announce message (e.g., a PTP ANNOUNCE message) to inform other nodes of its "Holdover-In-Specification" state (e.g., by setting clockClass=135 in the announce message, as defined in Table 3 below). Upon receiving the announce message, other nodes, such as T-BC-2 **220**, T-TSC-1 **222**, and T-TSC-2 **224** will be synchronized to the T-BC-1 **210** and aware of its Holdover-In-Specification state. Therefore, the nodes in the TS domain **20** can still be operated with an acceptable synchronization error as long as the T-BC-1 **210** stays in the Holdover-In-Specification state. In this way, even if the original grand master clock T-GM **200** is down, all other nodes in the TS domain **20** may be operated normally for a certain period of time, and if the T-GM **200** may come back online before the expiration of this period of time, then all the nodes may be synchronized to the T-GM **200** again and be operated normally.

(43) However, if the T-GM **200** cannot be restored during the Holdover-In-Specification period of T-BC-1 **210**, when a holdover budget at the T-BC-1 **210** is exhausted, or when the drifting error of the T-BC-1 **210** goes beyond the acceptable limit, T-BC-1 **210** may enter into the "Holdover-Out-Of-Specification" state, which indicates an unacceptable error level of its clock output, and may inform other nodes of its "Holdover-Out-Of-Specification" state via another announce message (e.g., by setting clockClass=165 in this announce message, as defined in Table 3 below).

(44) Further, as shown in FIG. **2**C, during the time period in which the T-GM **200** is down and the T-BC-1 **210** is in the Holdover-In-Specification state, the T-BC-2 **220** and its slave TS nodes T-TSC-1 **222** and T-TSC-2 **224** may also lose their time synchronization with the T-BC-1 **210**, for example, due to the power down of the T-BC-1 **210** or the failure of the communication link between the T-BC-1 **210** and the T-BC-2 **220**. Since there is no way for the T-BC-2 **220** to aware of to what extent the holdover budget at the T-BC-1 **210** is used or whether the holdover budget is exhausted or not, then the T-BC-2 **220** has to enter into the Holdover-Out-Of-Specification state immediately and directly although the performance of the clock output at the T-BC-2 **220** may still be acceptable.

(45) In other words, although the overall performance of the nodes of the TS domain **20** may be still acceptable, these nodes will consider themselves cannot provide acceptable services, resulting in a higher probability of service down and waste of resources.

(46) To improve the embodiment of FIG. **2**A-FIG. **2**C, another embodiment will be described with reference to FIG. **3**A-FIG. **3**G, in which a new HoldoverBudget TLV (Tag-Length-Value) of the

announce message is proposed to enable extended holdover time at nodes in a TS domain.

(47) An exemplary announce message or an exemplary PTP ANNOUNCE message is defined in Table 1 below.

(48) TABLE-US-00001 TABLE 1 Announce message fields Bits 7 6 5 4 3 2 1 0 Octets Offset header 34 0 originTimestamp 10 34 currentUtcOffset 2 44 reserved 1 46 grandmasterPriority1 1 47 grandmasterClockQuality 4 48 grandmasterPriority2 1 52 grandmasterIdentity 8 53 stepsRemoved 2 61 timeSource 1 63 header: It is defined in subclause 13.3.1 of IEEE 1588-2008. originTimestamp (Timestamp): The value of originTimestamp shall be 0 or an estimate no worse than ±1 s of the local time of the originating clock when the Announce message was transmitted. currentUtcOffset (Integer16): The value of currentUtcOffset shall be the value of the timePropertiesDS.currentUtcOffset member of the data set. grandmasterPriority1 (UInteger8): The value of grandmasterPriority1 shall be the value of the parentDS.grandmasterPriority1 member of the data set. grandmasterClockQuality (ClockQuality): The value of grandmasterClockQuality shall be the value of the parentDS.grandmasterClockQuality member of the data set. grandmasterPriority2 (UInteger8): The value of grandmasterPriority2 shall be the value of the parentDS.grandmasterPriority2 member of the data set. grandmasterIdentity (ClockIdentity): The value of grandmasterIdentity shall be the value of the parentDS.grandmasterIdentity member of the data set. stepsRemoved (UInteger16): The value of stepsRemoved shall be the value of currentDS.stepsRemoved of the data set of the clock issuing this message. timeSource (Enumeration8): The value of timeSource shall be the value of the timePropertiesDS.timeSource member of the data set.

(49) Further, the HoldoverBudget TLV mentioned above may be a suffix appended to the announce message, and it may be defined in Table 2 below.

(50) TABLE-US-00002 TABLE 2 HoldoverBudget TLV fields Bits 7 6 5 4 3 2 1 0 Octets Offset tlvType 2 0 lengthField 2 2 clockIdentity 8 4 holdoverBudget 4 12 tlvType (UInteger16): The value of tlvType shall be HOLDOVER_BUDGET, the value may be 4001. lengthField (UInteger16): The value of lengthField is 12. clockIdentity (ClockIdentity): The value of clockIdentity indicates a clock which is in the Holdover-In-Specification state. holdoverBudget (UInteger32): The value of holdoverBudget shall be the current budget at the clock for staying in the Holdover-In-Specification state.

(51) Further, an exemplary mapping between clockClass and states of TS node is given in Table 3 below.

(52) TABLE-US-00003 TABLE 3 Applicable clockClass values Phase/time traceability description clockClass T-GM connected to a PRTC 6 in locked mode T-GM in holdover, within 7 holdover specification T-BC in holdover, within 135 holdover specification T-GM in holdover, out of 140, 150, or 160, depending on a holdover specification frequency source to which T-GM is traceable. T-BC in holdover, out of 165 holdover specification

(53) By using this announce message with the HoldoverBudget TLV, a TS node may inform its slave TS nodes of its remaining budget for Holdover-In-Specification state, and each of the slave TS nodes may exploit this information for extending its own Holdover-In-Specification state if it loses time synchronization with its master node. In other words, with the knowledge of the remaining holdover budget, a TS node may not enter the Holdover-Out-Of-Specification state immediately and directly as shown in FIG. **2**C, and may continue the Holdover-In-Specification state instead, which obviously improves the robustness of the whole system against the temporary loss of time synchronization.

(54) To explain this holdover extension capability, an exemplary embodiment will be described in details with reference to FIG. **3**A-FIG. **3**G below.

(55) FIG. **3**A-FIG. **3**G are schematic diagrams illustrating a TS network **30** comprising multiple TS nodes according to another embodiment of the present disclosure.

(56) Referring to FIG. **3**A, the TS network or domain **30** may comprise multiple TS nodes, such as,

a grand master clock (e.g. T-GM **310**), four boundary clocks, T-BC-1 **320**, T-BC-2 **330**, T-BC-3 **340**, and T-BC-4 **350**, which are synchronized to each other. Further, there is also a primary reference time source (PRTS) **390** to which the T-GM **310** (and the whole TS network **30**) is synchronized, and it typically does not belong to the TS domain **30** according to the definition of "domain" above.

(57) As shown in FIG. **3**A, there is only one region comprising all the TS nodes of the TS domain **30**, or there is no region at all, depending on how the region is defined. By contrast, a multiple region scenario will be described with reference to FIG. **4**A-FIG. **4**F later. Further, the master/slave ports are not shown in FIG. **3**A-FIG. **3**G for simplicity, and one skilled in the art may readily understand that a TS node may be a slave TS node synchronized to an upper master TS node which is connected by a dotted line therebetween, and also is a master TS node synchronized to a lower slave TS node which is connected by a dotted line therebetween. Further, the T-GM **310** is synchronized to the PRTS **390**, which is the master TS node for the T-GM **310**.

(58) Please note that although FIG. **3**A-FIG. **3**G show only certain nodes and certain links between those nodes, the present disclosure is not limited thereto, and there may be additional nodes, less nodes, different nodes, and/or additional links, less links or different links therebetween. For example, a link may be present between T-BC-1 **320** and T-BC-3 **340** though it is not shown in FIG. **3**A-FIG. **3**G. For another example, an additional node such as a transparent clock T-TC may be present between the T-GM **310** and the T-BC-2 **330**.

(59) Referring back to FIG. **3**A, the TS domain **30** is operated in a locked state. In other words, each node of the TS domain **30** is synchronized or locked to its master node, and ultimately to the grand master node T-GM **310**, which is synchronized or locked to its master node PRTS **390**, which could be a GPS satellite in the embodiment shown in FIG. **3**A. Since all the nodes are locked to the T-GM **310** directly or indirectly, their clockClass will be identical to that of the grand master clock T-GM **310**, which is 6 in the embodiment of FIG. **3**A and according to the Table 3 above.

(60) Further, each TS node in the domain **30** will be pre-configured or assigned an initial holdover budget, for example, based on their own hardware capabilities (e.g. clock accuracy, oscillator level). As shown in FIG. **3**A, the T-GM **310**, T-BC-1 **320**, T-BC-2 **330**, T-BC-3 **340**, T-BC-4 **350** are pre-configured with holdover budgets of 800 ns, 300 ns, 350 ns, 400 ns, and 150 ns, respectively.

(61) Referring to FIG. **3**B, the T-GM **310** may lose its time synchronization with its master node PRTS **390**, and it changes its state from the "locked" state to the "Holdover-In-Specification" state by setting its clockClass to 7 and starting a holdover timer with its expiration time set as its initial holdover budget of 800 ns. Further, the T-GM **310** may transmit an announce message with the HoldoverBudget TLV to its slave TS nodes to inform them of the change of state and the remaining HoldoverBudget. In such a case, the remaining holdover budget is 800 ns, which will be appended to the announce message as shown in FIG. **3**B. Upon receiving the announce message with the holdover budget of 800 ns, each of the TS nodes, T-BC-1 **320**, T-BC-2 **330**, T-BC-3 **340**, and T-BC-4 **350**, will change their own clockClass to 7, and store the received holdover budget of 800 ns for future use.

(62) Referring to FIG. **3**C, the T-BC-2 **330** may detect loss of time synchronization with its master TS node, T-GM **310** (e.g. due to link failure between the T-GM **310** and the T-BC-2 **330**), and it may first determine whether its own state is "Holdover-In-Specification" state or not, for example, by checking its own clockClass. Since the clockClass at T-BC-2 **330** was 7 which indicates a Holdover-In-Specification state of its master TS node, T-GM **310**, the T-BC-2 **330** may change its clockClass from 7 to 135 to indicate it is still in the "Holdover-In-Specification" state but not a grand master clock in the domain **30**. After that, the T-BC-2 **330** may compare its previously received holdover budget of 800 ns and its pre-configured initial budget of 350 ns, and choose the minimum of those two budgets, i.e. 350 ns, as the expiration time for its own holdover timer to be started. Similar to FIG. **3**B, T-BC-2 **330** may transmit an announce message with a

HoldoverBudget of 350 ns to inform its slave TS nodes of the remaining budget for the part of the TS domain **30**, which comprises the T-BC-2 **330**, T-BC-3 **340**, and T-BC-4 **350**. Upon receiving the announce message from T-BC-2 **330**, the T-BC-3 **340** and T-BC-4 **350** may update its stored budgets of 800 ns which was previously received, as shown in FIG. **3**B, with the new budget of 350 ns as shown in FIG. **3**C. On the other hand, the T-BC-1 **320** is still synchronized to the T-GM **310**, and it receives another transmission of the announce message with a different budget of 750 ns as the time elapses. In other words, the T-GM **310** may transmit its remaining holdover budget to its slave TS nodes periodically, thereby updating its slave TS nodes with the latest remaining budget for the "Holdover-In-Specification" state. In such a case, T-BC-1 **320** may update its previously stored budget of 800 ns with the newly received budget of 750 ns as shown in FIG. **3**C.

(63) Referring to FIG. **3**D, the T-BC-3 **340** may detect loss of time synchronization with its master TS node, T-BC-2 **330**, and it may first determine whether its own state is "Holdover-In-Specification" state or not, for example, by checking its own clockClass. Since the clockClass at T-BC-3 **340** was 135 which indicates a Holdover-In-Specification state of its master TS node, T-BC-2 **330**, the T-BC-3 **340** may not change its clockClass since it is not the grand master clock in the domain **30** either. Further, the T-BC-3 **340** may compare its previously received holdover budget of 350 ns and its pre-configured initial budget of 400 ns, and choose the minimum of those two budgets, i.e. 350 ns, as the expiration time for its own holdover timer to be started. Similar to T-BC-2 **330** in FIG. **3**C, T-BC-3 **340** may transmit an announce message with a HoldoverBudget of 350 ns to inform its slave TS nodes of the remaining budget for the part of the TS domain **30** comprising the T-BC-3 **340**. Since no other TS node in this part of the TS domain **30** is shown in FIG. **3**D, then the announce message is not shown either. On the other hand, the T-BC-1 **320** is still synchronized to the T-GM **310**, and the T-BC-4 **350** is still synchronized to the T-BC-2 **330**. Therefore, they will update their stored budgets with the budgets of 700 ns and 300 ns which are received from T-GM **310** and T-BC-2 **330**, respectively, as shown in FIG. **3**D.

(64) Referring to FIG. **3**E, a relatively long period of time elapses, for example, 450 ns from the time instance shown in FIG. **3**D. At this time, T-GM **310** is still in the Holdover-In-Specification state due to its great holdover budget, and it will update the T-BC-1 **320** with its remaining budget of 250 ns by sending an announce message with a budget of 250 ns. On the other hand, the T-BC-1 **320** is still in the locked state and therefore its own initial budget is never used to be compared with the store budget which is received from the T-GM **310**.

(65) Further, the T-BC-2 **330**, T-BC-3 **340**, and T-BC-4 **350** use up their own holdover budgets, and have to change their states to the "Holdover-Out-Of-Specification" state as shown in FIG. **3**E. Since the T-BC-4 **350** is still locked to the T-BC-2 **330** and its own holdover timer was not started, its changing of state may be triggered by receiving an announce message from its master TS node T-BC-2 **330**, in which an indicator indicating a remaining budget of 0 and/or the change of state may be present.

(66) Referring to FIG. **3**F, after a further time period of 50 ns, the T-BC-2 **330** may resume its time synchronization with its master node T-GM **310**, and therefore it receives the announce message with the HoldoverBudget of 200 ns transmitted from the T-GM **310**. In such a case, the T-BC-2 **330** may change its clockClass from 165 back to 7 and its state from the "Holdover-Out-Of-Specification" state to the "locked" state. Further, since the T-BC-4 **350** is always locked to the T-BC-2 **330**, the T-BC-4 **350** may be triggered, by an announce message from the T-GM **310** or the T-BC-2 **330**, to change its clockClass from 165 back to 7 also. Further, the T-BC-2 **330** and the T-BC-4 **350** may restore their current holdover budgets to their initial holdover budgets of 350 ns and 150 ns, respectively.

(67) Referring to FIG. **3**G, after a yet further time period of 50 ns, the T-GM **310** may resume its time synchronization with its master node PRTS **390**, and therefore it may change its clockClass from 7 back to 6 and its state from the "Holdover-In-Specification" state to the "locked" state. Further, the T-GM **310** may transmit an announce message to its currently synchronized slave TS

nodes to inform them of the changing of the state, and these slave TS nodes may update their clockClass as well and restore their own budgets to the initial values. In some embodiments, these slave TS nodes may purge all the stored budget received from their master nodes.
(68) In this embodiment, it is clear that each TS node may continue to operate normally before its own holdover budget is exhausted and therefore extend its holdover time. By contrast, in the embodiment shown in FIG. **2**C, a TS node has to abort its normal operation even if its performance is still acceptable when it loses time synchronization with its master TS node which is in the Holdover-In-Specification state.
(69) Please note that the specific values (e.g., 800 ns, 300 ns, 350 ns, 6, 7, 135, 165, etc.), variables (e.g., clockClass, HoldoverBudget, etc.), messages (e.g. announce messages) are merely examples to explain the principle of the present disclosure, and the present disclosure is not limited thereto. In fact, in some other embodiments, different values, variables, messages, or any combinations thereof may be used. For example, in some other embodiments, the clockClass at the nodes may have a different definition than those described above, or different values of initial budgets may be assigned to the TS nodes. Further, in some other embodiments, a different message or a different field of the message may be used to indicate the holdover budget or any information from which the holdover budget may be derived. Further, in some embodiments, the variable “clockClass” may be omitted since the variable “HoldoverBudget” per se may implicitly indicate whether a node is in the Holdover-In-Specification state or the Holdover-Out-Of-Specification state, for example, by determining whether the HoldoverBudget is greater than zero or not.
(70) Next, another embodiment in which multiple regions are involved will be described in details with reference to FIG. **4**A-FIG. **4**F.
(71) FIG. **4**A-FIG. **4**F are schematic diagrams illustrating a TS network **40** comprising multiple TS nodes according to yet another embodiment of the present disclosure.
(72) Referring to FIG. **4**A, the TS network or domain **40** may comprise multiple TS nodes, such as, a grand master clock (e.g. T-GM **410**), two boundary clocks (T-BC-1 **420** and T-BC-2 **430**) and a slave-only ordinary clock (T-TSC-1 **440**), which are synchronized to each other. Further, there is also a primary reference time source (PRTS) **490** to which the T-GM **410** (and the whole TS network **40**) is synchronized, and it typically does not belong to the TS domain **40** according to the definition of “domain” above.
(73) As shown in FIG. **4**A, there are two regions in the domain **40**, one region comprising the T-GM **410** and the T-BC-1 **420**, and the other region comprising the T-BC-2 **430** and the T-TSC-1 **440**. However, please note that the present disclosure is not limited thereto and there may be more regions in a domain.
(74) Further, the master/slave ports are not shown in FIG. **4**A-FIG. **4**F for simplicity, and one skilled in the art may readily understand that a TS node may be a slave TS node synchronized to an upper or left master TS node which is connected by a dotted line therebetween, and a master TS node synchronized to a lower or right slave TS node which is connected by a dotted line therebetween. Further, the T-GM **410** is synchronized to the PRTS **490**, which is the master TS node for the T-GM **410**.
(75) Please note that although FIG. **4**A-FIG. **4**F show only certain nodes and certain links between those nodes, the present disclosure is not limited thereto, and there may be additional nodes, less nodes, different nodes, and/or additional links, less links or different links therebetween. For example, a link may be present between T-BC-1 **420** and T-TSC-1 **440** though it is not shown in FIG. **4**A-FIG. **4**F. For another example, an additional node such as a transparent clock T-TC may be present between the T-GM **410** and the T-BC-1 **420**.
(76) Referring back to FIG. **4**A, the TS domain **40** is operated in a locked state. In other words, each node of the TS domain **40** is synchronized or locked to its master node, and ultimately to the grand master node T-GM **410**, which is synchronized or locked to its master node PRTS **490**, which could be a GPS satellite in the embodiment shown in FIG. **4**A. Since all the nodes are locked to the

T-GM **410** directly or indirectly, their clockClass will be identical to that of the grand master clock T-GM **410**, which is 6 in the embodiment of FIG. **4**A and according to the Table 3 above.
(77) Further, each TS node in the domain **40** will be pre-configured or assigned an initial holdover budget, for example, based on their own hardware capabilities (e.g. clock accuracy, oscillator level). As shown in FIG. **4**A, the T-GM **410**, T-BC-1 **420**, T-BC-2 **430**, T-TSC-1 **440** are pre-configured with holdover budgets of 800 ns, 350 ns, 400 ns, and 150 ns, respectively.
(78) Referring to FIG. **4**B, the T-BC-1 **420** may lose its time synchronization with its master node the T-GM **410**, and it changes its state from the "locked" state to the "Holdover-In-Specification" state by setting its clockClass to 135 and starting a holdover timer with its expiration time set as its initial holdover budget of 350 ns. Further, the T-BC-1 **420** may transmit an announce message with the HoldoverBudget TLV to its slave TS nodes to inform them of the change of state and the remaining HoldoverBudget of 350 ns. Upon receiving the announce message with the holdover budget of 350 ns, each of the TS nodes, T-BC-2 **430** and T-TSC-1 **440**, will change their own clockClass to 135, and store the received holdover budget of 350 ns for future use.
(79) Referring to FIG. **4**C, the T-BC-2 **430** may detect loss of time synchronization with its master TS node, T-BC-1 **420**, and then it may first determine whether its master TS node was in the "Holdover-In-Specification" state or not, for example, by checking its own clockClass which is consistent with that of its master TS node. Since the clockClass at T-BC-2 **430** was 135 which indicates a Holdover-In-Specification state of its master TS node, T-BC-1 **420**, the T-BC-2 **430** may check whether the budget of 350 ns was received from a master TS node in a different region than itself, for example, by comparing the clock identity present in the received HoldoverBudget TLV with a pre-defined list of clock identifies within the same region as itself. In the embodiment of FIG. **4**C, the T-BC-2 **430** determines that the stored budget of 350 ns was received from T-BC-1 **420** which is located in a different region (e.g., the transport region) than that of itself (e.g., the radio region), and therefore the T-BC-2 **430** may calculate the sum of the received budget of 350 ns and its own pre-configured initial budget of 400 ns, i.e. 750 ns, as the expiration time for its own holdover timer to be started. Similar to T-BC-1 **420** in FIG. **4**B, T-BC-2 **430** may transmit an announce message with a HoldoverBudget of 750 ns to inform its slave TS nodes of the remaining budget for the part of the TS domain **40**, which comprises the T-TSC-1 **440** only in addition to the T-BC-2 **430** itself. Upon receiving the announce message from T-BC-2 **430**, the T-TSC-1 **440** may update its stored budgets of 350 ns which was previously received as shown in FIG. **4**B with the new budget of 750 ns as shown in FIG. **4**C.
(80) Referring to FIG. **4**D, the T-TSC-1 **440** may detect loss of time synchronization with its master TS node, T-BC-2 **430**, and then it may first determine whether its master TS node was in the "Holdover-In-Specification" state or not, for example, by checking its own clockClass which is consistent with that of its master TS node. Since the clockClass at T-TSC-1 **440** was 135 which indicates a Holdover-In-Specification state of its master TS node, T-BC-2 **430**, the T-TSC-1 **440** may check whether the budget of 750 ns was received from a master TS node in a different region than itself. In the embodiment of FIG. **4**D, the T-TSC-1 **440** determines that the stored budget of 750 ns was received from T-BC-2 **430** which is located in a same region (e.g., the radio region) as that of itself, and therefore the T-TSC-1 **440** may use the minimum of the received budget of 750 ns and its own pre-configured initial budget of 150 ns, i.e. 150 ns, as the expiration time for its own holdover timer to be started.
(81) Referring to FIG. **4**E, after a time period of 50 ns, the T-BC-2 **430** may resume its time synchronization with its master node T-BC-1 **420**, and therefore it receives the announce message with the HoldoverBudget of 200 ns transmitted from the T-BC-1 **420**. In such a case, the T-BC-2 **430** may be synchronized or locked to the T-BC-1 **420** and may restore its current holdover budget to its initial holdover budgets of 400 ns and update its saved holdover budget with 200 ns.
(82) Referring to FIG. **4**F, after another time interval of 50 ns, the T-BC-1 **420** may resume its time synchronization with its master node T-GM **410**, and therefore it may change its clockClass from

135 back to 6 and its state from the “Holdover-In-Specification” state to the “locked” state. Further, the T-BC-1 **420** may transmit an announce message to its currently synchronized slave TS nodes to inform them of the changing of the state, and these slave TS nodes may update their clockClass as well and restore their own budgets to the initial values. In some embodiments, these slave TS nodes may purge all the stored budget received from their master nodes.

(83) Similar to the embodiment shown in FIG. **3**A-FIG. **3**G, in the embodiment shown in FIG. **4**A-FIG. **4**F, each TS node may continue to operate normally before its own holdover budget is exhausted and therefore extend its holdover time. Even further, a TS node in a region may exploit the remaining budget of another TS node in another region, thereby further extending its holdover time. In fact, as shown in FIG. **4**A-FIG. **4**F, a user may never notice that there was a loss of time synchronization since the TS nodes may recover from the link failure before running out of their holdover time.

(84) FIG. **5** is a flow chart of an exemplary method **500** for enabling extended holdover time according to an embodiment of the present disclosure. The method **500** may be performed at a TS node (e.g., the TS nodes shown in FIG. **3**A-FIG. **3**G, or the TS nodes shown in FIG. **4**A-FIG. **4**F, or the device **600** shown in FIG. **6**) for enabling extended holdover time. The method **500** may comprise step S**510** and Step S**520**. However, the present disclosure is not limited thereto. In some other embodiments, the method **500** may comprise more steps, less steps, different steps or any combination therefore. Further the steps of the method **500** may be performed in a different order than that described herein. Further, in some embodiments, a step in the method **500** may be split into multiple sub-steps and performed by different entities, and/or multiple steps in the method **500** may be combined into a single step.

(85) The method **500** may begin at step S**510** where whether the TS node loses time synchronization with its master TS node or not is detected. At step S**520**, a first announce message (e.g. a PTP ANNOUNCE message) comprising a first indicator (e.g. a HoldoverBudget TLV) may be transmitted to one or more slave TS nodes in response to detecting that the TS node loses the time synchronization with its master TS node, wherein the first indicator indicates a holdover time budget for enabling extended holdover time at the one or more slave TS nodes.

(86) In some embodiments, after an initial transmission of the first announce message, one or more further transmissions of the first announce message may occur periodically with a value of the holdover time budget comprised in the first announce message being gradually reduced as the time elapses. In some embodiments, the method **500** may further comprise: entering into a “Holdover-In-Specification” state upon detecting that the TS node loses the time synchronization with its master TS node. In some embodiments, the first announce message may further comprise a second indicator (e.g., grandmasterClockQuality field in the PTP ANNOUNCE message) indicating that the TS node enters into the “Holdover-In-Specification” state. In some embodiments, the first indicator may indicate an initial holdover time budget which is pre-configured at the TS node. In some embodiments, upon detecting that the TS node loses the time synchronization with its master TS node, the method **500** may further comprise: starting a holdover timer with its expiration time set as the initial holdover time budget.

(87) In some embodiments, before detecting that the TS node loses the time synchronization with its master TS node, the method **500** may further comprise: receiving a second announce message which comprises a third indicator, the third indicator indicating a holdover time budget of the master TS node when the second announce message was transmitted; and storing the holdover time budget of the master TS node. In some embodiments, the first indicator may indicate the minimum of an initial holdover time budget which is pre-configured at the TS node and the stored holdover time budget of the master TS node if the master TS node belongs to a same TS region as that of the TS node. In some embodiments, upon detecting that the TS node loses the time synchronization with its master TS node, the method **500** may further comprise: starting a holdover timer with its expiration time set as the minimum of the initial holdover time budget which is pre-configured at

the TS node and the stored holdover time budget of the master TS node.
(88) In some embodiments, the first indicator may indicate the sum of an initial holdover time budget which is pre-configured at the TS node and the stored holdover time budget of the master TS node if the master TS node belongs to a TS region other than that of the TS node. In some embodiments, upon detecting that the TS node loses the time synchronization with its master TS node, the method **500** may further comprise: starting a holdover timer with its expiration time set as the sum of the initial holdover time budget which is pre-configured at the TS node and the stored holdover time budget of the master TS node.
(89) In some embodiments, the method **500** may further comprise: placing the TS node into a "Holdover-Out-Of-Specification" state upon the expiration of the holdover timer. In some embodiments, if multiple second announce messages are received from the master TS node, the step of storing the holdover time budget of the master TS node may comprise: overwriting the previously stored holdover time budget of the master TS node with the holdover time budget of the master TS node indicated in the last received second announce message.
(90) In some embodiments, in response to detecting that time synchronization with the master TS node which is synchronized to a primary reference time source (PRTS) or in the "Holdover-In-Specification" state is restored, the method **500** may further comprise: placing the TS node into the "locked" state; stopping the holdover timer; and transmitting an announce message to the one or more slave TS nodes to inform the restoration of the time synchronization.
(91) In some embodiments, in response to detecting that time synchronization with the master TS node which is synchronized to a primary reference time source (PRTS) is restored, the method further comprises: clearing the stored holdover time budget of the master TS node. In some embodiments, in response to detecting that time synchronization with the master TS node which is in the "Holdover-In-Specification" state is restored, the method further comprises: updating the holdover time budget of the master TS node when receiving the second announce message.
(92) In some embodiments, each of the TS nodes may be an IEEE 1588 Precision Time Protocol (PTP)-compliant node. In some embodiments, each of the announce messages may be a PTP-compliant ANNOUNCE message.
(93) FIG. **6** schematically shows an embodiment of an arrangement **600** which may be used in a TS node according to an embodiment of the present disclosure. Comprised in the arrangement **600** are a processing unit **606**, e.g., with a Digital Signal Processor (DSP) or a Central Processing Unit (CPU). The processing unit **606** may be a single unit or a plurality of units to perform different actions of procedures described herein. The arrangement **600** may also comprise an input unit **602** for receiving signals from other entities, and an output unit **604** for providing signal(s) to other entities. The input unit **602** and the output unit **604** may be arranged as an integrated entity or as separate entities.
(94) Furthermore, the arrangement **600** may comprise at least one computer program product **608** in the form of a non-volatile or volatile memory, e.g., an Electrically Erasable Programmable Read-Only Memory (EEPROM), a flash memory and/or a hard drive. The computer program product **608** comprises a computer program **610**, which comprises code/computer readable instructions, which when executed by the processing unit **606** in the arrangement **600** causes the arrangement **600** and/or the TS node in which it is comprised to perform the actions, e.g., of the procedure described earlier in conjunction with FIG. **2**A-FIG. **2**C, FIG. **3**A-FIG. **3**G, FIG. **4**A-FIG. **4**F, and FIG. **5** or any other variant.
(95) The computer program **610** may be configured as a computer program code structured in computer program modules **610**A-**610**B. Hence, in an exemplifying embodiment when the arrangement **600** is used in the TS node, the code in the computer program of the arrangement **600** includes: a detection module **610**A for detecting whether the TS node loses time synchronization with its master TS node or not. The code in the computer program further includes a transmission module **610**B for transmitting a first announce message comprising a first indicator to one or more

slave TS nodes in response to detecting that the TS node loses the time synchronization with its master TS node, wherein the first indicator indicates a holdover time budget for enabling extended holdover time at the one or more slave TS nodes.
(96) The computer program modules could essentially perform the actions of the flow illustrated in FIG. **2**A-FIG. **2**C, FIG. **3**A-FIG. **3**G, FIG. **4**A-FIG. **4**F, and FIG. **5**, to emulate the TS node. In other words, when the different computer program modules are executed in the processing unit **606**, they may correspond to different modules in the TS node.
(97) Although the code means in the embodiments disclosed above in conjunction with FIG. **6** are implemented as computer program modules which when executed in the processing unit causes the arrangement to perform the actions described above in conjunction with the figures mentioned above, at least one of the code means may in alternative embodiments be implemented at least partly as hardware circuits.
(98) The processor may be a single CPU (Central processing unit), but could also comprise two or more processing units. For example, the processor may include general purpose microprocessors; instruction set processors and/or related chips sets and/or special purpose microprocessors such as Application Specific Integrated Circuit (ASICs). The processor may also comprise board memory for caching purposes. The computer program may be carried by a computer program product connected to the processor. The computer program product may comprise a computer readable medium on which the computer program is stored. For example, the computer program product may be a flash memory, a Random-access memory (RAM), a Read-Only Memory (ROM), or an EEPROM, and the computer program modules described above could in alternative embodiments be distributed on different computer program products in the form of memories within the UE.
(99) The present disclosure is described above with reference to the embodiments thereof. However, those embodiments are provided just for illustrative purpose, rather than limiting the present disclosure. The scope of the disclosure is defined by the attached claims as well as equivalents thereof. Those skilled in the art can make various alternations and modifications without departing from the scope of the disclosure, which all fall into the scope of the disclosure.

Claims

1. A method at a time synchronization (TS) node for enabling extended holdover time, the method comprising: detecting whether the TS node loses time synchronization with its master TS node or not; transmitting a first announce message comprising a first indicator to one or more slave TS nodes in response to detecting that the TS node loses the time synchronization with its master TS node, wherein the first indicator indicates a holdover time budget to be used while the TS is in a holdover state, for enabling extended holdover time at the one or more slave TS nodes; and after an initial transmission of the first announce message, one or more further transmissions of the first announce message occurring periodically with a value of the holdover time budget comprised in the first announce message being gradually reduced as the time elapses.
2. The method of claim 1, further comprising: entering into a “Holdover-In-Specification” state upon detecting that the TS node loses the time synchronization with its master TS node.
3. The method of claim 2, wherein the first announce message further comprises a second indicator indicating that the TS node enters into the “Holdover-In-Specification” state.
4. The method of claim 1, wherein the first indicator indicates an initial holdover time budget which is pre-configured at the TS node.
5. The method of claim 4, wherein upon detecting that the TS node loses the time synchronization with its master TS node, the method further comprises: starting a holdover timer with its expiration time set as the initial holdover time budget.
6. The method of claim 5, further comprising: placing the TS node into a “Holdover-Out-Of-Specification” state upon the expiration of the holdover timer.

7. The method of claim 5, wherein in response to detecting that time synchronization with the master TS node which is synchronized to a primary reference time source (PRTS) or in the "Holdover-In-Specification" state is restored, the method further comprises: placing the TS node into the "locked" state; stopping the holdover timer; and transmitting an announce message to the one or more slave TS nodes to inform the restoration of the time synchronization.
8. The method of claim 7, wherein one or both of: in response to detecting that time synchronization with the master TS node which is synchronized to a primary reference time source (PRTS) is restored, the method further comprises: clearing the stored holdover time budget of the master TS node; and in response to detecting that time synchronization with the master TS node which is in the "Holdover-In-Specification" state is restored, the method further comprises: updating the holdover time budget of the master TS node when receiving the second announce message.
9. The method of claim 1, wherein before detecting that the TS node loses the time synchronization with its master TS node, the method further comprises: receiving a second announce message which comprises a third indicator, the third indicator indicating a holdover time budget of the master TS node when the second announce message was transmitted; and storing the holdover time budget of the master TS node.
10. The method of claim 9, wherein one or both of: the first indicator indicates the minimum of an initial holdover time budget which is pre-configured at the TS node and the stored holdover time budget of the master TS node if the master TS node belongs to a same TS region as that of the TS node; and the first indicator indicates the sum of an initial holdover time budget which is pre-configured at the TS node and the stored holdover time budget of the master TS node if the master TS node belongs to a TS region other than that of the TS node.
11. The method of claim 9, wherein one or both of: upon detecting that the TS node loses the time synchronization with its master TS node, the method further comprises: starting a holdover timer with its expiration time set as the minimum of the initial holdover time budget which is pre-configured at the TS node and the stored holdover time budget of the master TS node; and wherein upon detecting that the TS node loses the time synchronization with its master TS node, the method further comprises: starting a holdover timer with its expiration time set as the sum of the initial holdover time budget which is pre-configured at the TS node and the stored holdover time budget of the master TS node.
12. The method of claim 9, wherein if multiple second announce messages are received from the master TS node, the step of storing the holdover time budget of the master TS node comprises: overwriting the previously stored holdover time budget of the master TS node with the holdover time budget of the master TS node indicated in the last received second announce message.
13. The method of claim 1, wherein each of the TS nodes is an IEEE 1588 Precision Time Protocol (PTP)-compliant node.
14. The method of claim 13, wherein each of the announce messages is a PTP-compliant ANNOUNCE message.
15. A time synchronization (TS) node, comprising: a processor; a memory storing instructions which, when executed by the processor, cause the processor to perform: detecting whether the TS node loses time synchronization with its master TS node or not; transmitting a first announce message comprising a first indicator to one or more slave TS nodes in response to detecting that the TS node loses the time synchronization with its master TS node, wherein the first indicator indicates a holdover time budget to be used while the TS is in a holdover state, for enabling extended holdover time at the one or more slave TS nodes; and after an initial transmission of the first announce message, one or more further transmissions of the first announce message occurring periodically with a value of the holdover time budget comprised in the first announce message being gradually reduced as the time elapses.